

One Way Travel

CITS3001 Advanced Algorithms — Time limit: 3s

Travis lives in a town of N places joined by *one way* roads, which makes it a confusing place to drive. It is quite possible to drive somewhere and then find you cannot return without driving the wrong way up a one way street. Worse, the police are vigilant: every time Travis drives the wrong way along a one way road he collects exactly one demerit point. Driving a road in its legal direction is free.

Travis must drive from his origin O to his destination D , and then drive back from D to O . The two legs are independent — on each leg he may use any roads he likes, in either direction, as often as he likes. Work out the minimum total number of demerit points he collects.

Input

The first line contains three integers N , O and D ($1 \leq N \leq 10^5$, $0 \leq O < N$, $0 \leq D < N$): the number of places, the origin and the destination. Places are numbered 0 to $N - 1$.

Each of the next N lines describes the roads leading *out* of one place. Line i (for $i = 0, 1, \dots, N - 1$) begins with an integer k_i , the number of one way roads out of place i , followed by the k_i distinct places j ($0 \leq j < N$, $j \neq i$) that those roads lead to. The total number of roads $\sum_i k_i$ is at most $2 \cdot 10^5$.

If the directions of the roads are ignored, the network is connected, so the round trip is always possible.

Output

Output a single integer: the minimum total number of demerit points Travis collects driving from O to D and back again.

Note

Some networks are 10^5 places deep. A recursive depth-first search will need its recursion limit raised (in Python, `sys.setrecursionlimit`), or should be rewritten with an explicit stack.

Sample Input 1

```
5 0 4
2 1 2
1 3
1 3
1 4
0
```

Sample Output 1

```
3
```

Sample Input 2

```
4 0 3
1 1
1 2
2 0 3
0
```

Sample Output 2

```
1
```